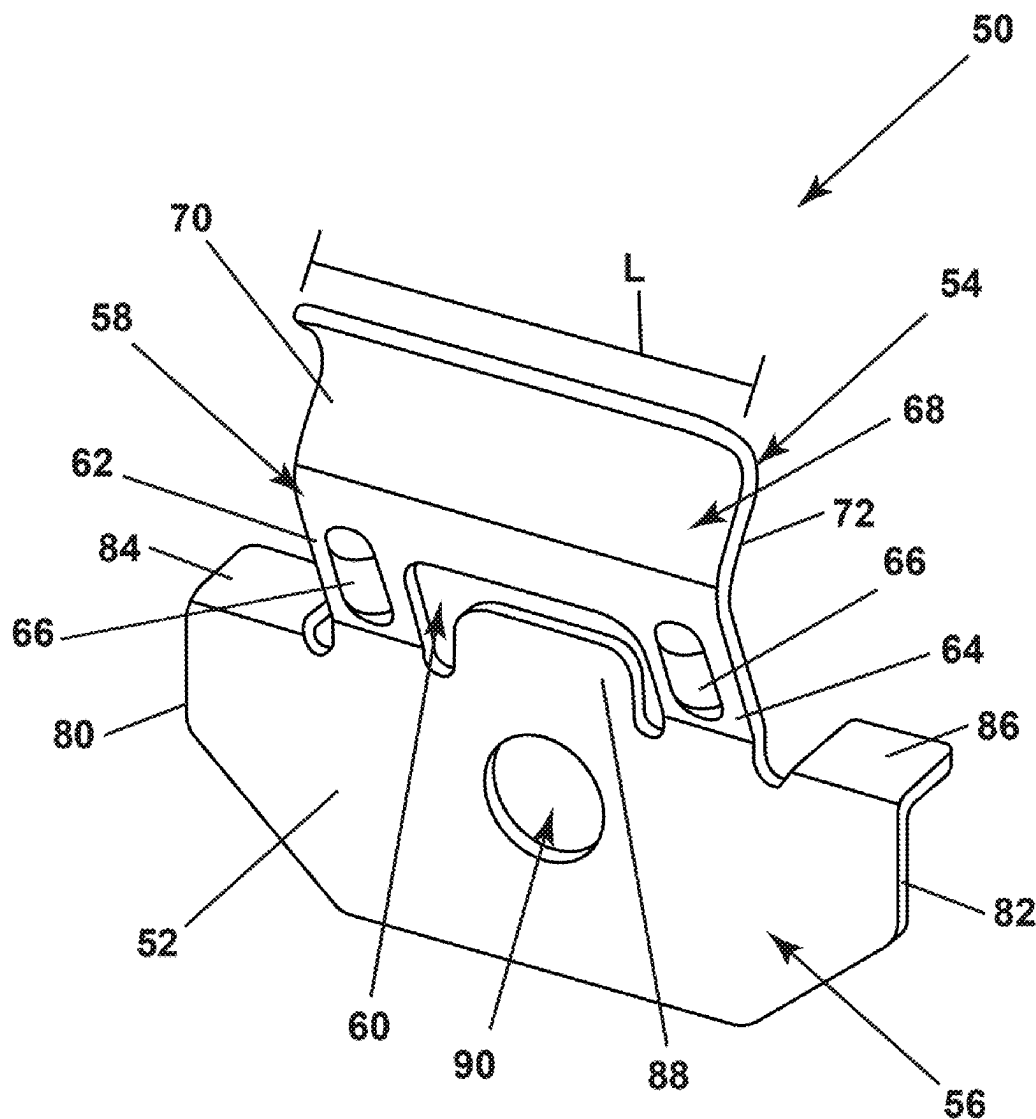




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AUER et al.(10) **Pub. No.: US 2025/0271143 A1**(43) **Pub. Date: Aug. 28, 2025**(54) **COOKING APPLIANCE WITH CLIP**(52) **U.S. Cl.**(71) Applicant: **WHIRLPOOL CORPORATION,**
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SHINDE, PUNE (IN)(57) **ABSTRACT**(21) Appl. No.: **18/587,393**(22) Filed: **Feb. 26, 2024****Publication Classification**(51) **Int. Cl.**
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A cooking appliance with a cooking area such as a stovetop or an oven can be utilized to prepare and cook food items. The cooking appliance includes a chassis forming a frame for the cooking appliance. A console couples to the chassis for operating the cooking appliance, such as including knobs or other controls for operating the stovetop or the oven. A clip removably couples the console to the chassis. The clip has a body including a spring finger for securing the console against the chassis between the spring finger and the chassis by a spring force.



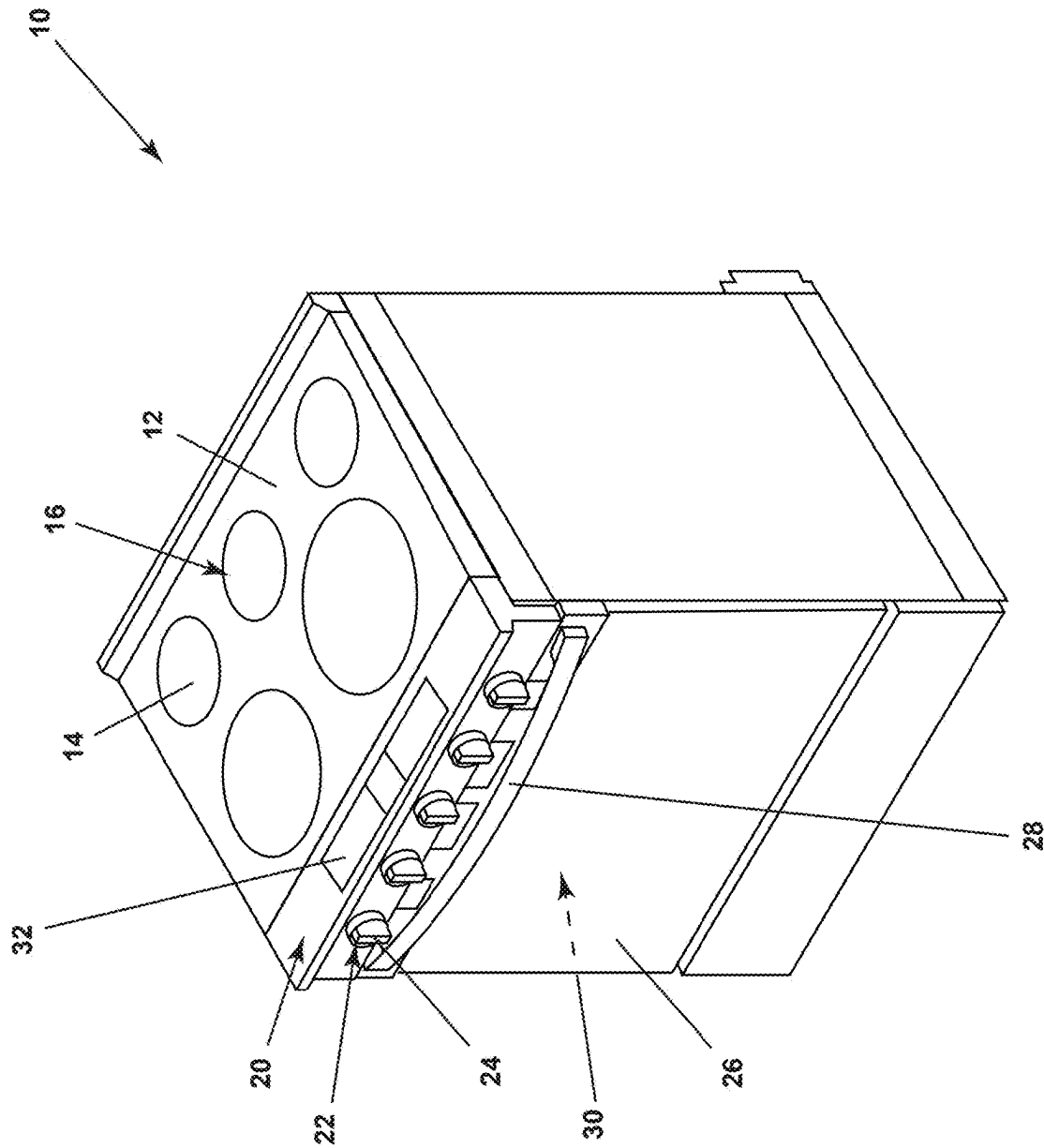


FIG. 1

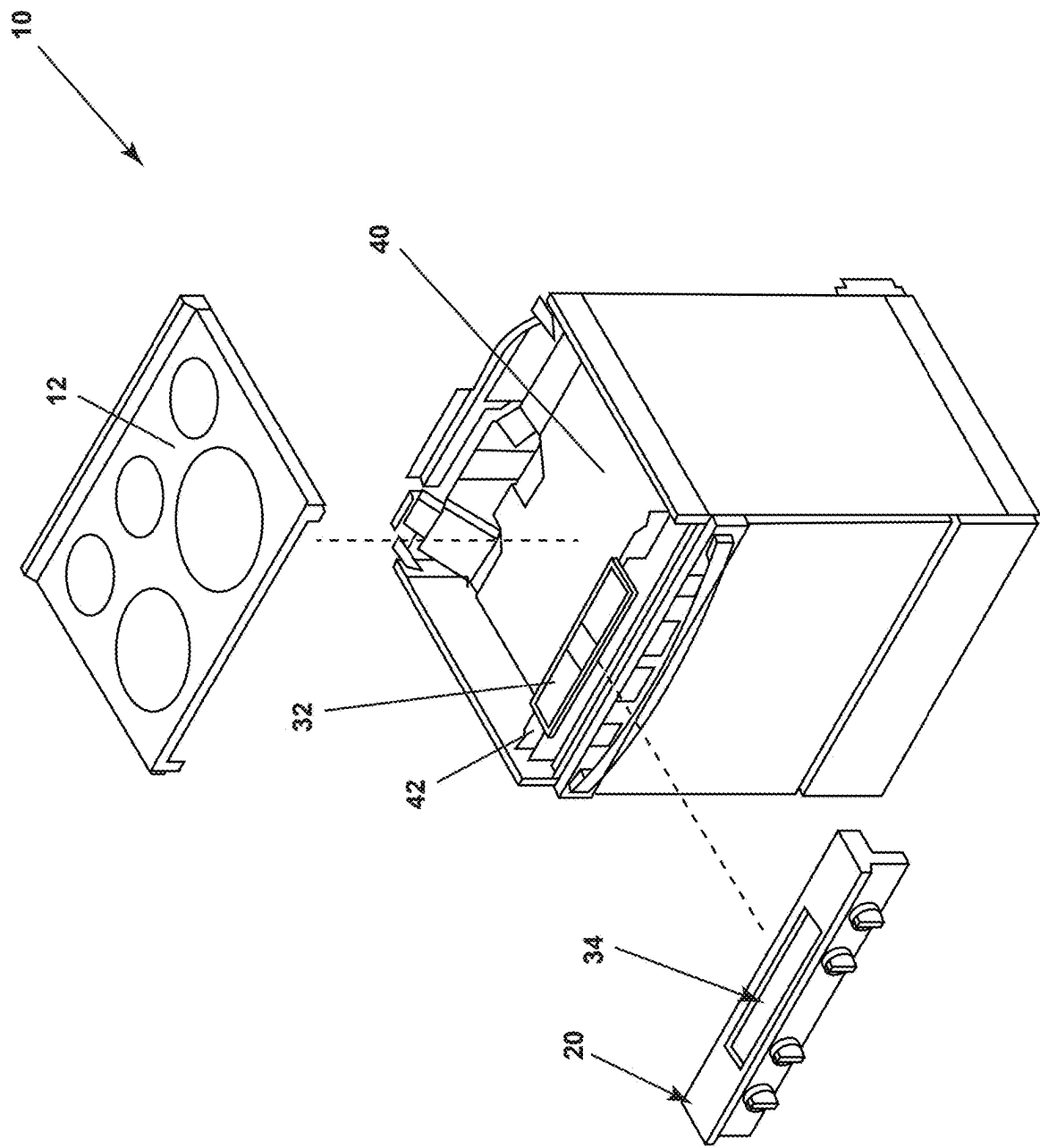


FIG. 2

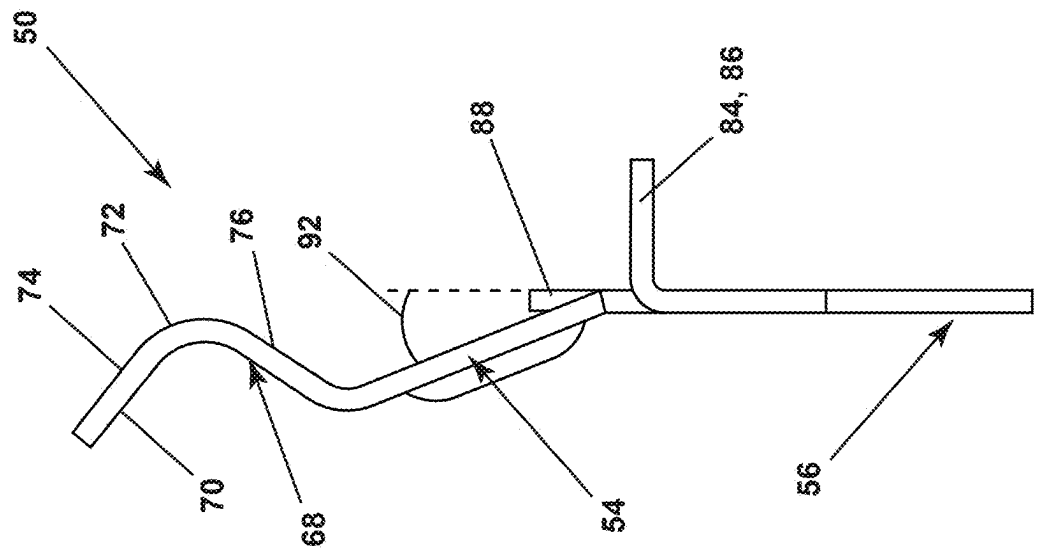


FIG. 4

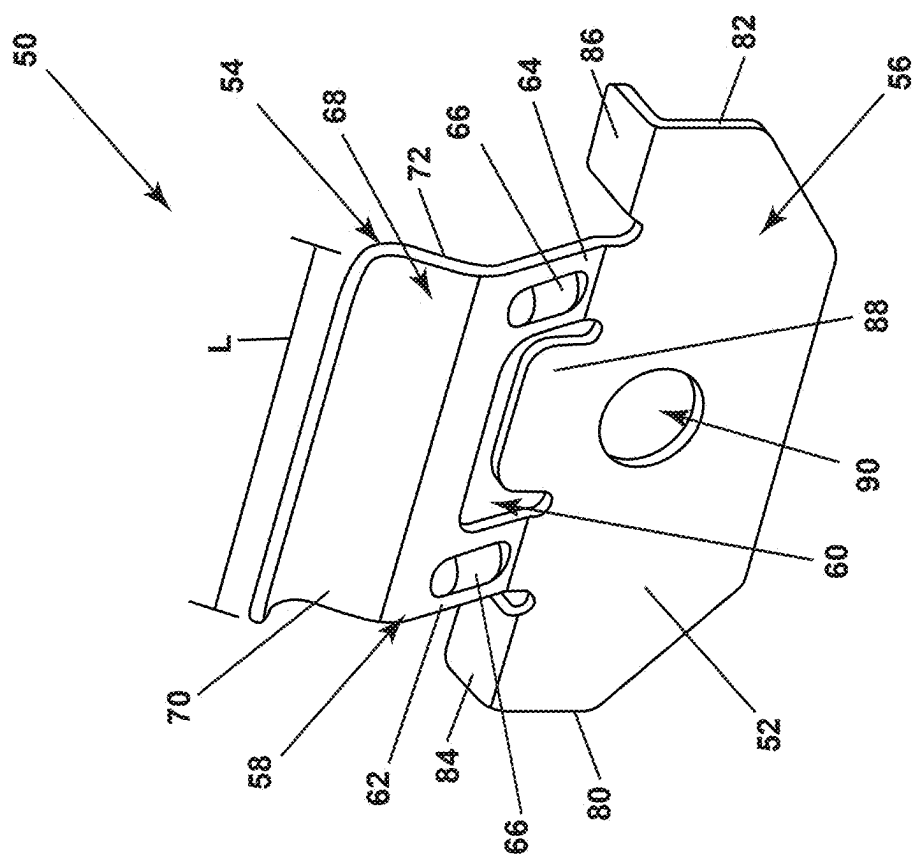


FIG. 3

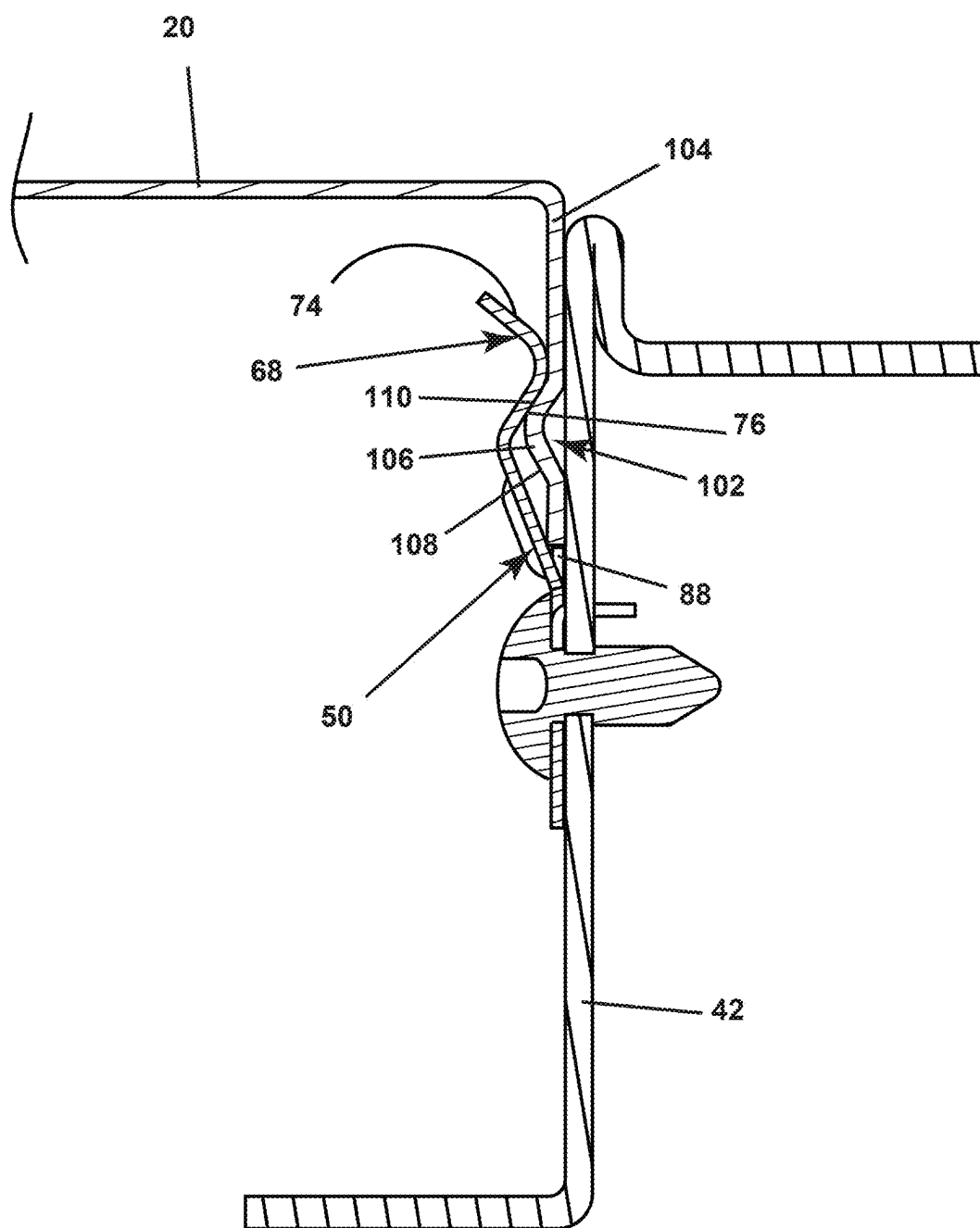


FIG. 6

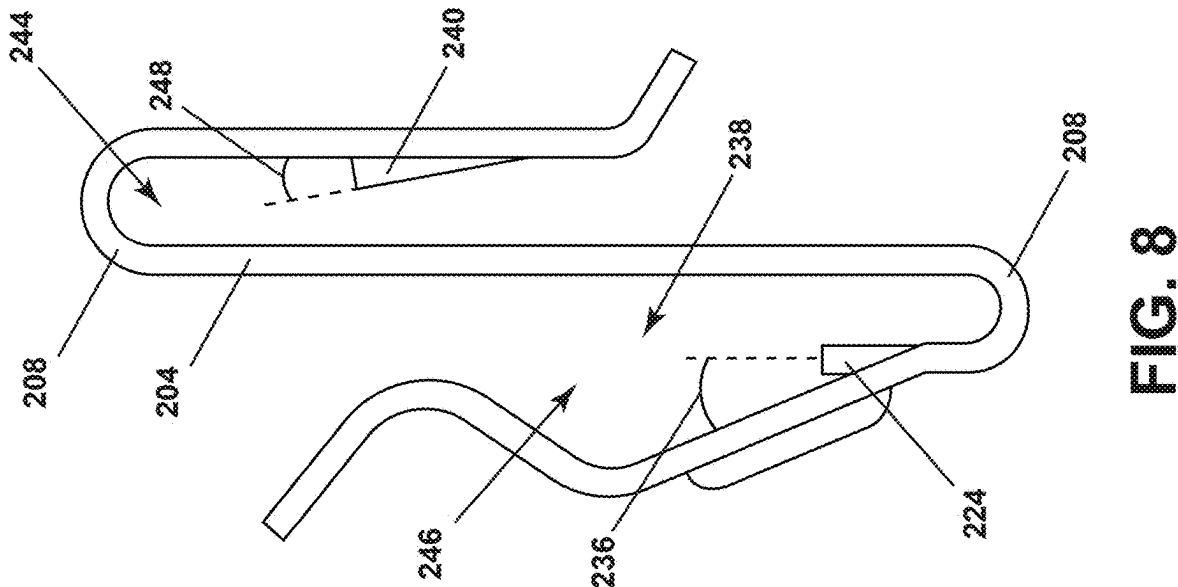


FIG. 8

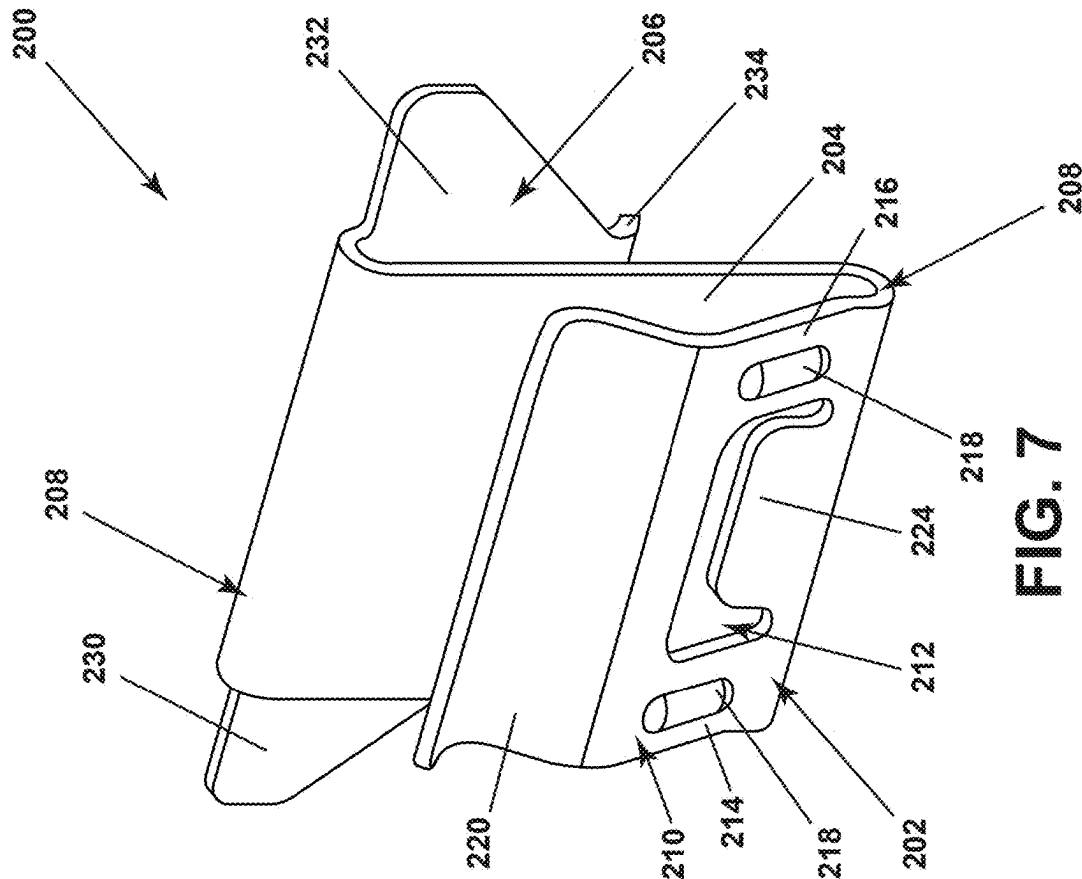


FIG. 7

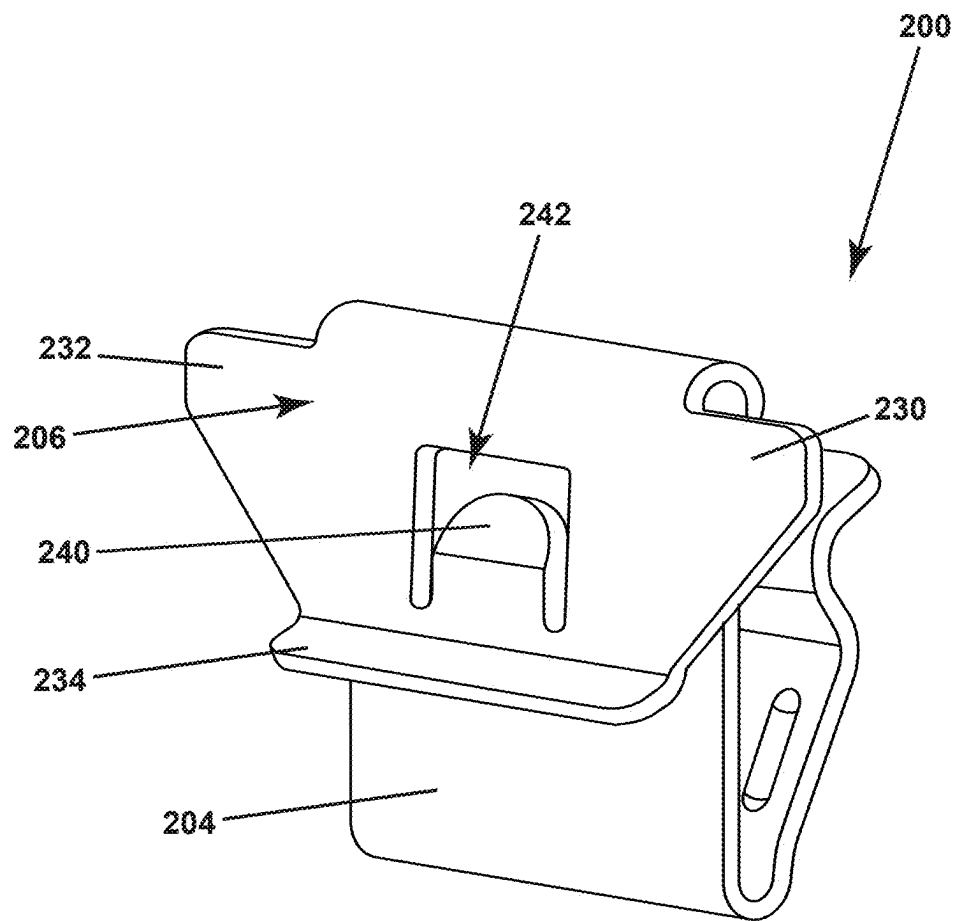
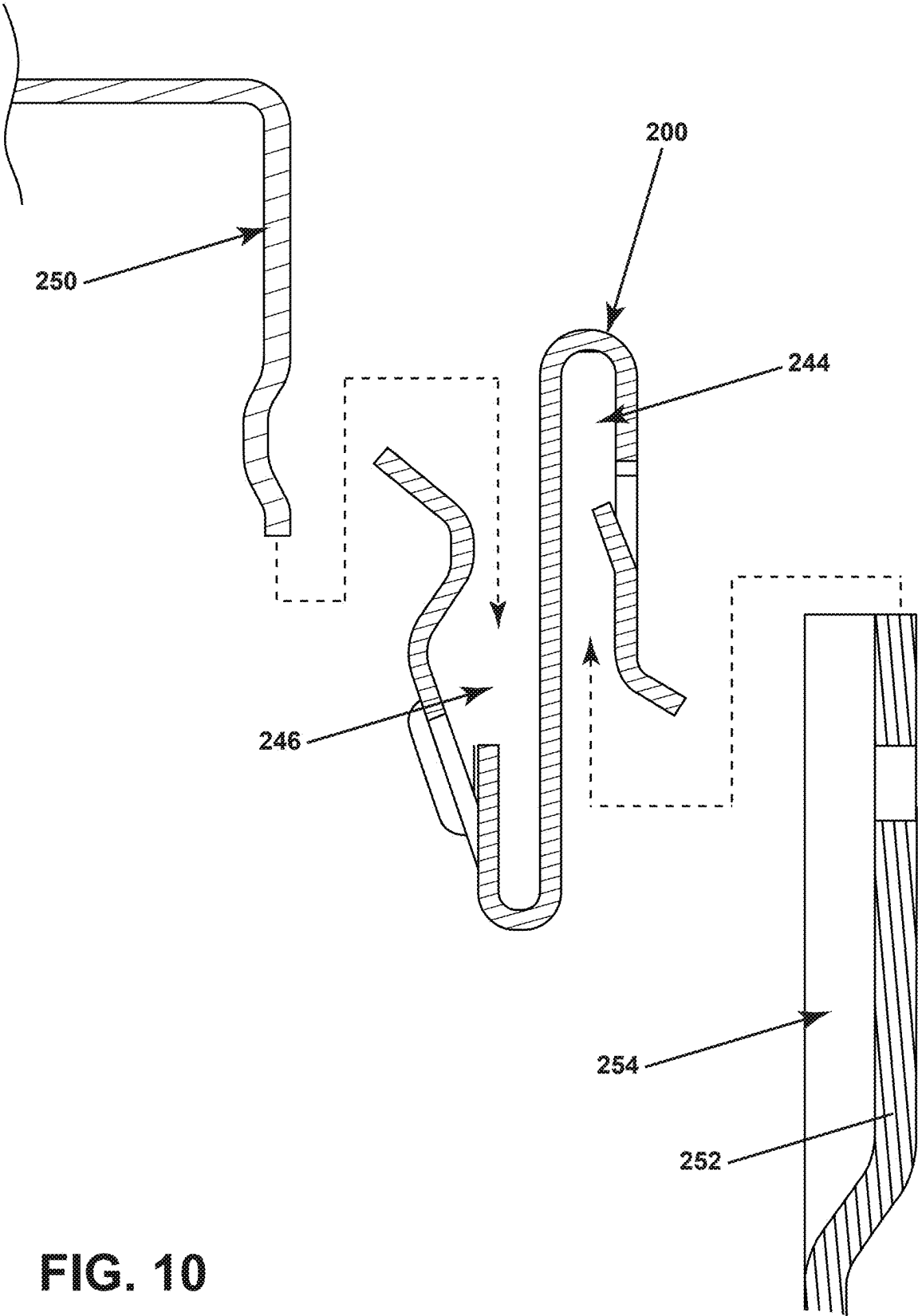


FIG. 9



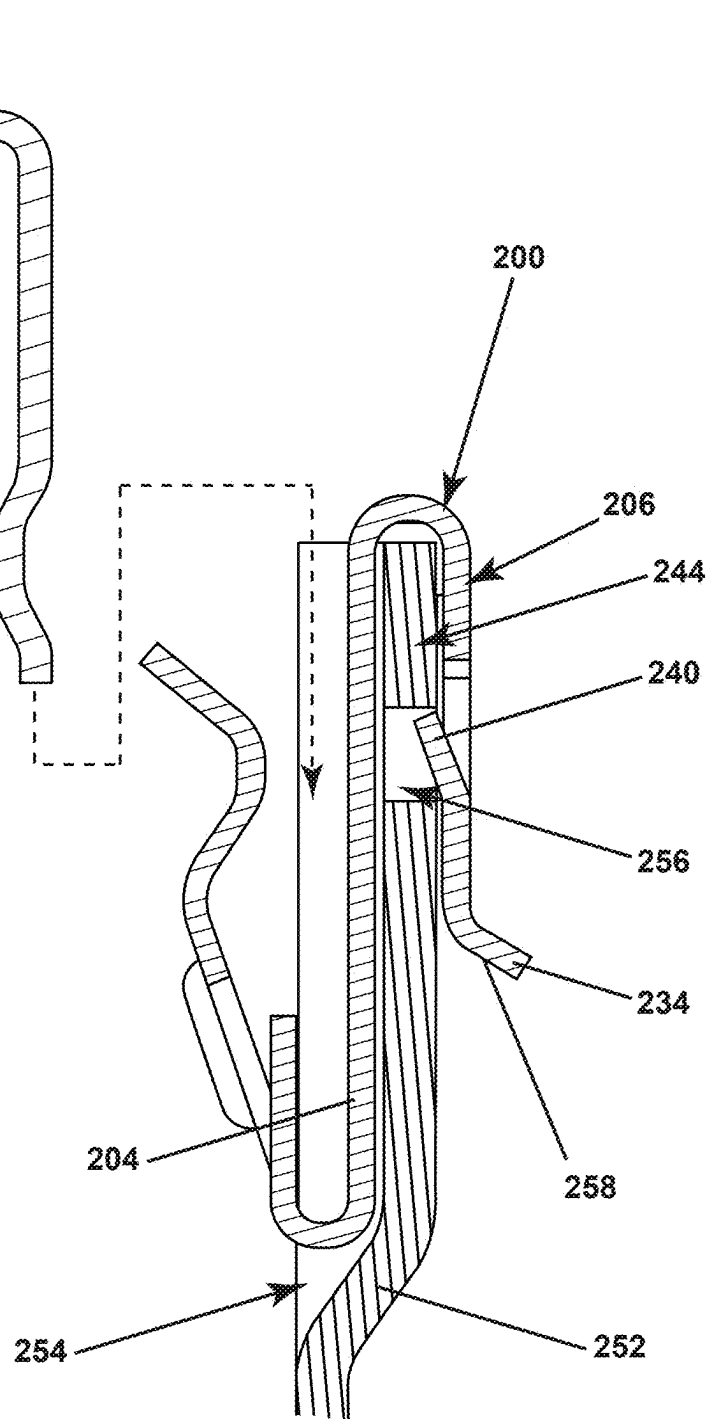


FIG. 11

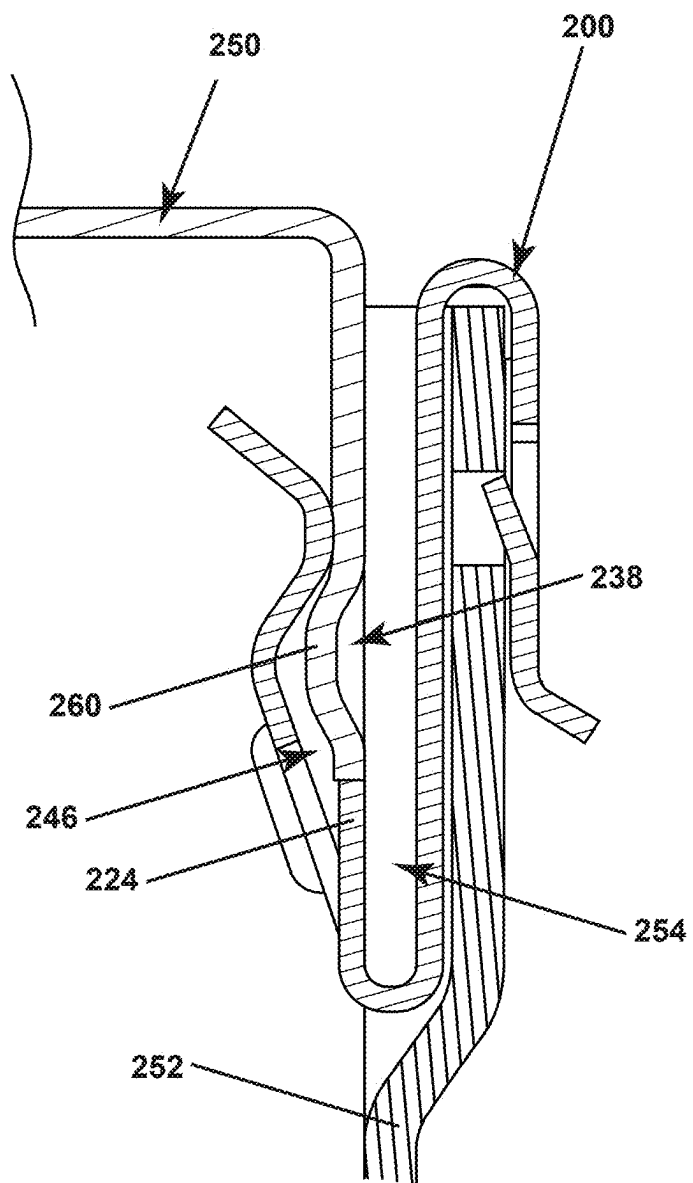


FIG. 12

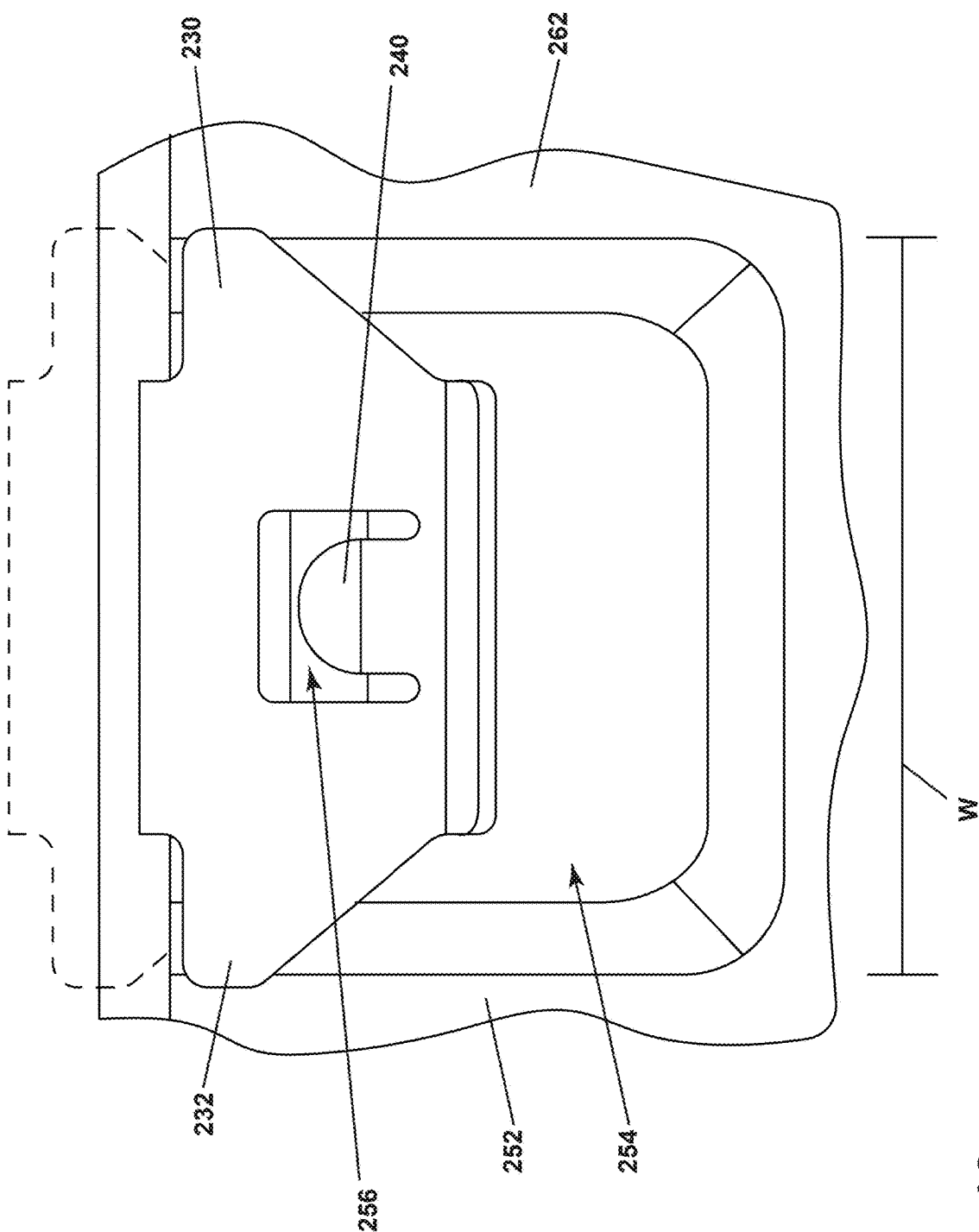


FIG. 13

COOKING APPLIANCE WITH CLIP

FIELD OF THE DISCLOSURE

[0001] This disclosure relates to a household appliance, such as a cooking appliance like a stove, an oven, or a combination thereof, and, to a clip for securing a console to a chassis of the cooking appliance.

BACKGROUND

[0002] Stovetops, such as gas stovetops, are utilized for cooking food or other activities that utilize heating materials with heightened temperatures. Stovetops that utilize combustible fuels, such as gas or charcoal, can utilize buttons, knobs, or other controls to operate burning of the combustible fuels, such as by controlling flow rates of a fluid or gas fuel to vary flame size and temperature output.

SUMMARY

[0003] In one aspect, the disclosure relates to a cooking appliance with a cooking area, the cooking appliance comprising: a chassis; a console coupled to the chassis for operating the cooking appliance; and a clip coupling the console to the chassis, the clip having a body including a spring finger; wherein the spring finger secures the console against the chassis between the spring finger and the chassis by a spring force.

[0004] In another aspect, the disclosure relates to a clip for coupling a console to a chassis for a cooking appliance, the clip comprising: a body; a spring finger portion including a spring finger; and an extension spacing the spring finger portion from a remainder of the body; wherein the extension is arranged at an angle relative to a remainder of the body that is greater than or equal to 1-degree and less than or equal to 45-degrees.

BRIEF DESCRIPTION OF THE DRAWINGS

[0005] In the drawings:

[0006] FIG. 1 depicts a cooking appliance with a cooktop and a console, in accordance with an aspect of the present disclosure.

[0007] FIG. 2 depicts the cooking appliance of FIG. 1 having the cooktop and the console exploded from the cooking appliance, exposing a chassis including an attachment wall for attaching to the console, in accordance with an aspect of the present disclosure.

[0008] FIG. 3 depicts a clip used to couple the console to the attachment wall of FIG. 2, in accordance with an aspect of the present disclosure.

[0009] FIG. 4 depicts a side view of the clip of FIG. 3, showing an angled orientation of a spring finger portion relative to an attachment portion, in accordance with an aspect of the present disclosure.

[0010] FIG. 5 depicts a side, sectional view of the clip of FIG. 3 coupled to the attachment wall of FIG. 2 by a fastener and having the console of FIG. 2 positioned above the clip to prepare for attachment of the console to the clip, in accordance with an aspect of the present disclosure.

[0011] FIG. 6 depicts another side, sectional view of the assembly of FIG. 5 having the console attached to the chassis at the attachment wall with the clip, in accordance with an aspect of the present disclosure.

[0012] FIG. 7 depicts a clip used to couple a console to a chassis for a cooking appliance, in accordance with an aspect of the present disclosure.

[0013] FIG. 8 depicts a side view of the clip of FIG. 7 showing an angled orientation of a spring finger portion relative to an attachment portion, in accordance with an aspect of the present disclosure.

[0014] FIG. 9 depicts a rear perspective view of the clip of FIG. 7, illustrating a tab and first and second wings utilized to ensure proper positioning of the clip, in accordance with an aspect of the present disclosure.

[0015] FIG. 10 depicts an exploded, sectional view showing a chassis and a console exploded from the clip of FIG. 7, in accordance with an aspect of the present disclosure.

[0016] FIG. 11 depicts a partially exploded, sectional view of the assembly of FIG. 10, having the clip coupled to the chassis, in accordance with an aspect of the present disclosure.

[0017] FIG. 12 depicts a sectional view of the assembly of FIG. 11, having the clip coupling the console to the chassis, in accordance with an aspect of the present disclosure.

[0018] FIG. 13 depicts a rear view of the assembly of FIG. 12, illustrating a depression in the console attachment wall and how the first and second wings of FIG. 9 ensure proper positioning of the clip relative to the attachment wall at the depression, in accordance with an aspect of the present disclosure.

DETAILED DESCRIPTION

[0019] As may be appreciated, based on the disclosure, there exists a need in the art for an assembly facilitating attachment of a console to a chassis for a cooking appliance. The disclosure herein includes a clip that permits a user or manufacturer to easily align and attach a console unit to the chassis for the cooking appliance, as well as receive feedback that the proper attachment has been made. The force required to attach the console to the clip can be less than the force required to remove the console from the clip to facilitate installation, while preventing unintended removal of the console from the clip.

[0020] Referring to FIG. 1, a cooking appliance 10 includes a cooktop 12 having a set of burners 14 defining an exterior cooking area 16. The set of burners 14 in a non-limiting example can be gas burners, such as for burning synthetic gas, natural gas, propane, or other flammable liquids, gasses, or fuels in further non-limiting examples to cook or otherwise heat cooking items such as food items within the exterior cooking area 16. While five burners are shown for the set of burners 14, any number of burners are contemplated, including one or more.

[0021] A console 20 is positioned at a front of the cooktop 12. A set of controllers 22, depicted as knobs 24, are utilized to control operation of the set of burners 14, such as by turning the set of burners 14 on and off, as well as controlling flame size or heat produced with the set of burners 14. In the illustrated example, each of the set of burners 14 has a corresponding one of the knobs 24. A door 26 is positioned below the console 20 and includes a handle 28 to selectively open and close the door 26 for accessing an interior cooking area 30, such as an oven. The set of controllers 22 can further include a control panel 32, which can be provided on the console 20. In a non-limiting example, the control panel 32 can be utilized to control operation of the interior cooking area 30, such as by controlling the temperature of the interior

cooking area 30. While control of the cooking appliance 10 is illustrated by way of the knobs 24 and the control panel 32, it is contemplated that any suitable system, feature, or controller can be incorporated into the console 20 to operate the cooking appliance 10.

[0022] Referring to FIG. 2, the cooktop 12 and the console 20 are exploded from the cooking appliance 10 to expose a chassis 40. The control panel 32 is coupled to the chassis 40, exposing an opening 34 in the console 20 when the console 20 is removed from the cooking appliance 10. In an alternate, non-limiting example, it is contemplated that the control panel 32 is dedicated to the console 20 and attaches to the chassis 40 via the console 20.

[0023] An attachment wall 42 couples to or forms a portion of the chassis 40. To couple the console 20 to the cooking appliance 10, the console 20 couples to the chassis 40 at the attachment wall 42. The attachment wall 42 can be hidden by the cooktop 12 and the console 20 when the cooking appliance is fully assembled (see FIG. 1).

[0024] FIG. 3 shows a clip 50 that can be utilized for coupling a console to a chassis for a cooking appliance, such as for coupling the console 20 to the attachment wall 42 for the cooking appliance 10 of FIG. 2, for example. In a non-limiting example, the clip 50 can be for an electric cooking appliance, while a gas or other appliance or fuel type is contemplated. The clip 50 includes a body 52 having a spring finger portion 54 and a chassis attachment portion 56. The spring finger portion 54 can be used to secure the clip 50 to the console 20 and the chassis attachment portion 56 can be used to secure the clip 50 to the chassis 40 at attachment wall 42 (see FIG. 2).

[0025] The spring finger portion 54 extends from the chassis attachment portion 56 at an extension portion 58. An opening 60 is provided at least partially within the extension portion 58, separating the extension portion 58 into a first extension 62 and a second extension 64. The first and second extensions 62, 64 can permit flexing or bending of the spring finger portion 54 relative to the chassis attachment portion 56 when acted upon by a large enough force. The size, material, thickness, and spacing among the first and second extensions 62, 64 can define a spring force for flexion or bending of the spring finger portion 54. A protuberance 66 can be provided in or on each of the first and second extensions 62, 64 to reinforce the first and second extensions 62, 64 against weakening when bending relative to the chassis attachment portion 56. A spring finger 68 extends from the extension portion 58 opposite of the chassis attachment portion 56. The spring finger 68 includes an exterior surface 70 and a contact surface 72 opposite the exterior surface 70. The spring finger 68 includes a common curved or bent shape extending along a length (L).

[0026] The chassis attachment portion 56 extends between a first side 80 and a second side 82, having a greater width than that of the spring finger portion 54. A first alignment tab 84 extends from the chassis attachment portion 56 at the first side 80 and a second alignment tab 86 extends from the chassis attachment portion 56 at the second side 82. A stop 88 extends from the chassis attachment portion 56 toward the spring finger portion 54 at the opening 60 and can be sized to fit within the opening 60. A fastener opening 90 extends through the chassis attachment portion 56 and can be positioned centrally between the first side 80 and the second side 82.

[0027] FIG. 4 illustrates the clip 50 from a side view, better illustrating the orientation of the spring finger portion 54 relative to the chassis attachment portion 56. The extension portion 58 of the spring finger portion 54 is arranged at an angle 92 relative to the chassis attachment portion 56. The angle 92 can be greater than 1-degree) (1° and less than or equal to 25-degrees) (25° in a non-limiting example, while any angle between 1-degree) (1° and 90-degrees) (90° is contemplated, such that the spring finger portion 54 is oriented at a non-zero angle relative to the chassis attachment portion 56. In another non-limiting example, the angle can be 21-degrees) (21°). The spring finger 68 includes a concave curvature along the exterior surface 70 and a convex curvature along the contact surface 72, further defining an upper contact surface 74 and a lower contact surface 76 resultant of the curvature of the spring finger 68. The first and second alignment tabs 84, 86 can be bent at a 90-degree angle relative to the remainder of the chassis attachment portion 56. The stop 88 can be aligned with the chassis attachment portion 56.

[0028] Referring to FIG. 5 the clip 50 provides for attaching the console 20 to the chassis 40 at the attachment wall 42. Initially, the clip 50 can be coupled to the attachment wall 42. A fastener hole 96 can be provided in the attachment wall 42 and is sized similar to the fastener opening 90 within the clip 50. A fastener 98 couples the clip 50 to the attachment wall 42 through the fastener opening 90 at the fastener hole 96.

[0029] Slots 100 (shown in broken line) extend through the attachment wall 42. The first and second alignment tabs 84, 86 can insert into the slots 100 during attachment of the clip 50 to the attachment wall 42. Insertion of the first and second alignment tabs 84, 86 into the slots 100 ensures proper alignment and positioning of the clip 50 relative to the attachment wall 42. Furthermore, inserting the first and second alignment tabs 84, 86 into the slots 100 prevents unintended rotation of the clip 50 when securing the clip 50 to the attachment wall 42. For example, the fastener 98 may be a threaded fastener which can otherwise cause an unintended rotation of a clip relative to the attachment wall 42 without having the benefit of the first and second alignment tabs 84, 86 inserted into the slots 100.

[0030] The angled orientation of the spring finger portion 54 relative to the chassis attachment portion 56 results in a cavity 102 arranged between the spring finger 68 and the attachment wall 42 when the clip 50 couples to the attachment wall 42. The console 20 can include a console wall 104 with a protrusion 106. The protrusion 106 can be sized to fit within the cavity 102. The protrusion 106 can have a curved cross-sectional shape, similar to that of the spring finger 68.

[0031] Referring to FIG. 6, attachment of the console 20 to the attachment wall 42 is achieved by inserting the console wall 104 into the cavity 102. The protrusion 106 is pushed down against the spring finger 68, causing the spring finger 68 to bend and move away from the attachment wall 42 to permit room for the protrusion 106 to pass into the cavity 102. More specifically, the protrusion 106 is pushed against the upper contact surface 74 to cause the spring finger 68 to bend away from the attachment wall 42 until enough room is provided for the protrusion 106 to pass into the cavity 102. The console wall 104 can be fully inserted into the cavity 102 when it abuts the stop 88. The elongated shape for the spring finger 68 defined by the length (L) (FIG. 3) can form a lap joint with the protrusion 106, among the

parallel surfaces in contact with one another. Such a lap joint prevents a rotational offset between the spring finger 68 and the protrusion 106.

[0032] Once the protrusion 106 is inserted into the cavity 102, the spring finger 68 returns toward the attachment wall 42 utilizing a spring force created by the angled orientation of the spring finger portion 54 and by bending of the spring finger portion away from the attachment wall 42. The spring force generated by the spring finger 68 creates a pressure against the console wall 104 between the clip 50 and the attachment wall 42 that secures the console wall 104 within the cavity 102 at the protrusion 106. The elongated shape of the spring finger 68 having the length (L) (see FIG. 3) can be complementary with an elongated shape for the protrusion 106. The elongated shape of the spring finger 68 and a similar elongated shape for the protrusion 106 can prevent rotational misalignment between the spring finger 68 and the protrusion 106, which can resist or prevent rotation of the console 20 relative to the clip 50. In this way, proper alignment between the clip 50 and the console 20 is achieved and misalignment is prevented.

[0033] When the console 20 is properly inserted at the clip 50, the spring finger 68 can generate an audible 'click' or other noise to indicate to the installer or user that attachment has been achieved. Such a click or noise can be generated by the spring finger 68 contacting the console 20 due to the spring force returning the spring finger 68 toward its original position after insertion of the protrusion 106 into the cavity 102. Additionally, vibrations generated by the spring finger 68 can provide haptic feedback to the user to indicate that proper installation has occurred.

[0034] In a non-limiting example, spring finger 68 can be shaped and sized such that the force required for insertion of the protrusion 106 into the cavity 102 is greater than the force required for removal of the protrusion 106 from the cavity 102. This can be achieved with the particular sizes and shapes for the spring finger 68 and the protrusion 106. For example, the upper contact surface 74 can be arranged at a lesser angle than that of the lower contact surface 76, relative to the console wall 104, such that a greater force is required to displace the spring finger 68 when pressing against the upper contact surface 74 than when pulling against the lower contact surface 76. In this way, a lesser force is required to attach the console 20 to the clip 50 than that required to remove the console 20 from the clip 50. In another non-limiting example, the shape or size for the protrusion 106 can determine the amount of force required to attach or remove the console 20 from the clip 50. For example, a lower surface 108 for the protrusion 106 can have a lesser angle of incline than an upper surface 110, thereby requiring less force to insert the protrusion 106 into the cavity 102 than to remove the protrusion 106 from the cavity 102. In a non-limiting example, the force required to attach the console wall 104 to the clip 50 can be 56.87 Newtons (N) and the force required to remove the console wall 104 from the clip 50 can be 65 N. In another non-limiting example, the force required to attach the console wall 104 to the clip 50 can be at least 3 N less than the force required to remove the console wall 104 from the clip 50.

[0035] Referring to FIG. 7, a clip 200 is provided that can be used to couple a console to a chassis for a cooking appliance, such as the console 20 and chassis 40 of FIGS. 1-2, for example. In a non-limiting example, the clip 200 can be for a gas cooking appliance, while electric or other fuel

types are contemplated. The clip 200 includes a spring finger portion 202, a center wall 204, and a chassis attachment portion 206, with the center wall 204 separating the spring finger portion 202 from the chassis attachment portion 206. The spring finger portion 202 and the chassis attachment portion 206 each connect to the center wall 204 at a curved junction 208, which can have a 180-degree turn between connected portions.

[0036] The spring finger portion 202 includes an extension portion 210 extending from the curved junction 208. An opening 212 is provided at least partially within the extension portion 210, defining a first extension 214 and a second extension 216. A protuberance 218 can be provided on one or both of the first and second extensions 214, 216. A spring finger 220 extends from the extension portion 210 opposite of the curved junction 208. The spring finger 220 includes a curved or bent shape similar to that discussed with respect to the spring finger 68 of FIGS. 3 and 4. A stop 224 extends from the curved junction 208 toward the opening 212. The chassis attachment portion 206 includes a first wing 230 and a second wing 232 and terminates at a lip 234 opposite of the curved junction 208.

[0037] Referring to FIG. 8, the stop 224 extends from the curved junction 208 parallel to the center wall 204, while the spring finger portion 202 is arranged at an angle 236 offset from parallel to the center wall 204 at the extension portion 210. The angle 236 can be greater than or equal to 1-degree) (1° and less than or equal to 25-degrees) (25°, while any angle greater than or equal to 1-degree) (1° and less than or equal to 90-degrees) (90° is contemplated. In one example, the angle 236 can be 21-degrees) (21°. The angled arrangement of the extension portion 210 defines a cavity 238 between the extension portion 210, the spring finger 220, and the center wall 204. A tab 240 extends at an angle 248 from the chassis attachment portion 206 toward the center wall 204. A first attachment region 244 can be defined as the area between the center wall 204 and the chassis attachment portion 206 and a second attachment region 246 can be defined as the area between the center wall 204 and the spring finger portion 202.

[0038] FIG. 9 shows a rear view of the clip 200 with the tab 240 is at least partially positioned within and extending from an extension opening 242 formed in the chassis attachment portion 206. The lip 234 extends between the first wing 230 and the second wing 232 and extends in a direction away from the center wall 204.

[0039] Referring to FIG. 10, a schematic section view illustrates the clip 200 separated from a console 250 and an attachment wall 252 or chassis. To attach the console 250 to the attachment wall 252 utilizing the clip 200, the attachment wall 252 is inserted into the first attachment region 244 and the console 250 is inserted into the second attachment region 246. The attachment wall 252 can further include a depression 254, such that at least a portion of the attachment wall 252 forms a valley or basin that is unaligned with the remainder of the attachment wall 252.

[0040] Referring to FIG. 11, the clip 200 can attach to the attachment wall 252 at the depression 254 by inserting the attachment wall 252 into the first attachment region 244, or by placing the clip 200 onto the attachment wall 252 and positioning the attachment wall 252 into the first attachment region 244. Insertion of the attachment wall 252 into the first attachment region 244 depresses the tab 240 to permit the attachment wall 252 to extend to the curved junction 208 and

fully insert into the first attachment region 244. A tab opening 256 is provided in the attachment wall 252 to receive the tab 240 when the attachment wall 252 has been fully inserted into the first attachment region 244. When the tab 240 inserts into the tab opening 256, an audible “click” or other audible feedback is generated, instructing the installer or user that a proper attachment has occurred. Additionally, the snap of the tab 240 inserting into the tab opening 256 provides haptic feedback to the user to inform the user that proper attachment has occurred. The tab 240 secures the clip 200 to the attachment wall 252 at the tab opening 256, where the tab 240 prevents removal of the clip 200 from the first attachment region 244. Removal of the clip 200 can be achieved by pushing against a bottom 258 of the lip 234. More specifically, pushing the lip 234 can bend the chassis attachment portion 206 of the clip 200, removing the tab 240 from the tab opening 256, permitting removal of the clip 200 from the attachment wall 252 at the first attachment region 244. After attachment, at least a portion of the center wall 204 can position within the depression 254.

[0041] Referring to FIG. 12, the clip 200 can attach to the console 250 by inserting the console 250 into the second attachment region 246. The console 250 does not extend into the depression 254 due to the length of the console 250 being longer than that of the depression 254, thereby preventing the console 250 from entering the depression 254. During attachment, when the console 250 is moving into the second attachment region 246, a protrusion 260 on the console 250 forces the spring finger 220 in a direction away from the center wall 204, providing room for the protrusion 260 to pass by the spring finger 220 and insert into the cavity 238. The spring finger 220 can form a lap joint with the protrusion 260, which can prevent a rotational offset among the spring finger 220 and the protrusion. After the protrusion 260 moves into the cavity 238, the spring finger 220 returns toward its original position because of a spring force generated by the spring finger 220 bending away from the attachment wall 252. The protrusion 260 secures the console 250 within the second attachment region 246 at the spring finger 220. The spring force imparted by the spring finger 220 against the console 250 presses the console 250 against the attachment wall 252 to secure the console 250. When the protrusion 260 is secured within the cavity 238, the console 250 can rest against the stop 224.

[0042] When the console 250 is properly inserted, the spring finger 220 can generate an audible ‘click’ or other noise to indicate to the installer or user that attachment has been achieved. Such a click or noise can be generated by the spring finger 220 contacting the console 250 due to the spring force returning the spring finger 220 toward its original position. Additionally, vibrations generated by the spring finger 220 can provide haptic feedback to the user to indicate that proper installation has occurred.

[0043] Removal of the console 250 from the clip 200 can be achieved by depressing the spring finger 220 away from the console 250, permitting the protrusion 260 to exit the cavity 238. The console 250 can then be removed from the second attachment region 246.

[0044] Referring to FIG. 13, the depression 254 is more readily appreciable in the attachment wall 252, and the depression 254 extends out of the page from the rear-view of FIG. 13. As can be appreciated, the span for the first and second wings 230, 232 is greater than a width (W) for the depression 254, thereby preventing improper insertion of the

clip 200. Due to the span of the first and second wings 230, 232, the clip 200 is only insertable against a rear surface 262 of the attachment wall 252, thereby ensuring proper positioning of the clip 200 at the attachment wall 252. If a user or installer attempts to attach the clip 200 with the first and second wings 230, 232 extending into the depression 254, the span of the first and second wings 230, 232 being greater than the width (W) of the depression 254 will prevent insertion of the attachment wall 252 into the first attachment region 244, as the first and second wings 230, 232 will contact the depression 254. Such an interference is shown in broken line in FIG. 13, preventing further insertion into the depression 254 by the first and second wings 230, 232 against the attachment wall 252. The tab 240 is not permitted to reach the tab opening 256. In this way, proper orientation and attachment of the clip 200 relative to the attachment wall 252 is ensured.

[0045] Benefits can be realized by use of the clip 50, 200 for attaching a console 20, 250 to a chassis 40 or attachment wall 42, 252 of the chassis 40. The clip 50, 200 as described herein provides for improvements for attachment of a console 20, 250 to a chassis 40. The clip 50, 200 provides feedback to the user or installer, such as audible or haptic feedback, to let the user or installer know that a proper attachment has been achieved. The insertion force required to install the console 20, 250 to the clip 50, 200 can be less than the force required to remove the console 20, 250 from the clip 50, 200 which facilitates installation, while reducing the possibility of unintended detachment of the console 20, 250 at the clip 50, 200. The shape for the spring finger 68, 220 on the clip 50, 200 forms a lap joint with the protrusion 106, 260 on the console 20, 250 which prevents a rotational offset for the clip 50, 200 relative to the console 20, 250. Similarly, the first and second alignment tabs 84, 86 ensure proper installation position, and prevent rotational misalignment between the clip 50 and the attachment wall 42 of the chassis 40. Furthermore, the first and second wings 230, 232 prevent incorrect orientation of the clip 200 relative to the attachment wall 252. These features ensure proper positioning and alignment of the clip 50, 200, and therefore proper alignment and positioning of the console 20, 250 relative to the chassis 40.

[0046] Further benefits can include a positive removable lock for the clip at both the console 20, 250 and the chassis 40 or attachment wall 42, 252. Specifically, the tab 240 inserting into the tab opening 256 on the attachment wall 252 ensures proper positioning of the clip 200, as well as preventing unintended separation of the clip 200 from the attachment wall 252. The lip 234 permits positive removal of the clip 200 from the attachment wall 252.

[0047] Further still, the design of the clip 50, 200 improves assembly and installation, reducing manufacture and finishing costs and time, as well as improving serviceability of the cooking appliance 10.

[0048] Many other possible embodiments and configurations in addition to that shown in the above figures are contemplated by the present disclosure. To the extent not already described, the different features and structures of the various embodiments can be used in combination with each other. That one feature cannot be illustrated in all the embodiments is not meant to be construed that it cannot be but is done for brevity of description. Thus, the various features of the different embodiments can be mixed and matched to form new embodiments, whether or not the new

embodiments are expressly described. Moreover, while “a set of” various elements have been described, it will be understood that “a set” can include any number of the respective elements, including only one element. Combinations or permutations of features described herein are covered by this disclosure.

[0049] This written description uses examples to disclose embodiments of the disclosure, including the best mode, and to enable any person skilled in the art to practice embodiments of the disclosure, including making and using any devices or systems and performing any incorporated methods. The patentable scope of the disclosure is defined by the claims, and can include other examples that occur to those skilled in the art. Such other examples are intended to be within the scope of the claims if they have structural elements that do not differ from the literal language of the claims, or if they include equivalent structural elements with insubstantial differences from the literal languages of the claims.

[0050] Further aspects are provided by the subject matter of the following clauses:

[0051] A cooking appliance with a cooking area, the cooking appliance comprising: a chassis; a console coupled to the chassis for operating the cooking appliance; and a clip coupling the console to the chassis, the clip having a body including a spring finger; wherein the spring finger secures the console against the chassis between the spring finger and the chassis by a spring force.

[0052] The cooking appliance of any preceding clause, wherein the clip further comprises a chassis attachment portion for coupling the clip to the chassis.

[0053] The cooking appliance of any preceding clause, wherein the spring finger is offset from the chassis attachment portion by an angle.

[0054] The cooking appliance of any preceding clause, wherein the angle is greater than or equal to 1-degree and less than or equal to 25-degrees.

[0055] The cooking appliance of any preceding clause, further comprising a tab provided on the chassis attachment portion.

[0056] The cooking appliance of any preceding clause, wherein the chassis further comprises a tab opening for receiving the tab when the clip is attached to the chassis.

[0057] The cooking appliance of any preceding clause, wherein the chassis attachment portion further comprises a lip.

[0058] The cooking appliance of any preceding clause, wherein the clip further comprises a center wall spacing the chassis attachment portion from the spring finger.

[0059] The cooking appliance of any preceding clause, wherein the clip further comprises a first attachment region defined between the center wall and the chassis attachment portion, and wherein the chassis is received within the first attachment region when the clip is coupled to the chassis.

[0060] The cooking appliance of any preceding clause, wherein the clip further comprises a second attachment region defined between the center wall and the spring finger, and wherein the console is received within the second attachment region when the clip is coupled to the console.

[0061] The cooking appliance of any preceding clause, wherein the center wall meets each of the chassis attachment portion and the spring finger at a curved junction.

[0062] The cooking appliance of any preceding clause, wherein the spring finger is shaped such that a force required

to attach the console to the clip is lesser than a force required to remove the console from the clip.

[0063] The cooking appliance of any preceding clause, wherein the console further comprises a console wall having a bump sized to insert at the spring finger.

[0064] The cooking appliance of any preceding clause, wherein the console further comprises a protrusion sized to be received by the spring finger.

[0065] The cooking appliance of any preceding clause, wherein the protrusion meets the spring finger at a lap joint.

[0066] The cooking appliance of any preceding clause, further comprising a pair of extensions spacing the spring finger from a remainder of the body.

[0067] The cooking appliance of any preceding clause, wherein the spring finger is flexible relative to the body at the pair of extensions to generate the spring force.

[0068] The cooking appliance of any preceding clause, further comprising a protuberance provided on each of the pair of extensions.

[0069] The cooking appliance of any preceding clause, further comprising an opening in the body between the pair of extensions.

[0070] The cooking appliance of any preceding clause, further comprising a stop extending from the body toward the opening.

[0071] The cooking appliance of any preceding clause, wherein the body further comprises a fastener opening for coupling the clip to the chassis with a fastener.

[0072] The cooking appliance of any preceding clause, wherein the chassis comprises a fastener hole for receiving the fastener from the fastener opening.

[0073] The cooking appliance of any preceding clause, further comprising an alignment tab extending perpendicular from the body.

[0074] The cooking appliance of any preceding clause, wherein the chassis further comprises a slot for receiving the alignment tab when the clip is coupled to the chassis.

[0075] The cooking appliance of any preceding clause, further comprising a pair of wings extending from the body to define a span among the pair of wings.

[0076] The cooking appliance of any preceding clause, wherein the chassis includes a depression having a width that is less than the span among the pair of wings.

[0077] A clip for coupling a console to a chassis for a cooking appliance, the clip comprising: a body; a spring finger portion including a spring finger; and an extension spacing the spring finger portion from a remainder of the body; wherein the extension is arranged at an angle relative to a remainder of the body that is greater than or equal to 1-degree and less than or equal to 45-degrees.

[0078] The clip of any preceding clause, wherein the extension is one of a pair of extensions spacing the spring finger portion from the remainder of the body, defining an opening between the pair of extensions.

[0079] The clip of any preceding clause, wherein the spring finger portion is flexible relative to the body at the pair of extensions to generate the spring force.

[0080] The clip of any preceding clause, further comprising a protuberance provided on each of the pair of extensions.

[0081] The clip of any preceding clause, further comprising an opening in the body between the pair of extensions.

[0082] The clip of any preceding clause, further comprising a stop extending from the body toward the opening.

[0083] The clip of any preceding clause, further comprising an alignment tab extending perpendicular from the body.

[0084] The clip of any preceding clause, further comprising a pair of wings extending from the body to define a span among the pair of wings.

[0085] The clip of any preceding clause, wherein the clip further comprises a chassis attachment portion.

[0086] The clip of any preceding clause, wherein the clip further comprises a center wall spacing the chassis attachment portion from the spring finger portion.

[0087] The clip of any preceding clause, wherein the center wall meets each of the chassis attachment portion and the spring finger portion at a curved junction to define a first attachment region between the center wall and the spring finger portion and a second attachment region between the center wall and the chassis attachment portion.

[0088] The clip of any preceding clause, further comprising a tab provided on the chassis attachment portion.

[0089] The clip of any preceding clause, further comprising a lip extending from the chassis attachment portion.

What is claimed is:

1. A cooking appliance with a cooking area, the cooking appliance comprising:

a chassis;

a console coupled to the chassis for operating the cooking appliance; and

a clip coupling the console to the chassis, the clip having a body including a spring finger;

wherein the spring finger secures the console against the chassis between the spring finger and the chassis by a spring force.

2. The cooking appliance of claim 1, wherein the clip further comprises a chassis attachment portion for coupling the clip to the chassis.

3. The cooking appliance of claim 2, wherein the spring finger is offset from the chassis attachment portion by an angle.

4. The cooking appliance of claim 3, wherein the angle is greater than or equal to 1-degree and less than or equal to 25-degrees.

5. The cooking appliance of claim 2, wherein the clip further comprises a center wall spacing the chassis attachment portion from the spring finger.

6. The cooking appliance of claim 5, wherein the clip further comprises a first attachment region defined between the center wall and the chassis attachment portion, and wherein the chassis is received within the first attachment region when the clip is coupled to the chassis.

7. The cooking appliance of claim 6, wherein the clip further comprises a second attachment region defined between the center wall and the spring finger, and wherein

the console is received within the second attachment region when the clip is coupled to the console.

8. The cooking appliance of claim 1, wherein the spring finger is shaped such that a force required to attach the console to the clip is lesser than a force required to remove the console from the clip.

9. The cooking appliance of claim 1, wherein the console further comprises a protrusion sized to be received by the spring finger.

10. The cooking appliance of claim 9, wherein the protrusion meets the spring finger at a lap joint.

11. The cooking appliance of claim 1, further comprising a pair of extensions spacing the spring finger from a remainder of the body.

12. The cooking appliance of claim 1, further comprising an alignment tab extending perpendicular from the body.

13. The cooking appliance of claim 12, wherein the chassis further comprises a slot receiving the alignment tab when the clip is coupled to the chassis.

14. The cooking appliance of claim 1, further comprising a pair of wings extending from the body to define a span among the pair of wings.

15. The cooking appliance of claim 14, wherein the chassis includes a depression having a width that is less than the span among the pair of wings.

16. A clip for coupling a console to a chassis for a cooking appliance, the clip comprising:

a body;

a spring finger portion including a spring finger; and

an extension spacing the spring finger portion from a remainder of the body;

wherein the extension is arranged at an angle relative to a remainder of the body that is greater than or equal to 1-degree and less than or equal to 45-degrees.

17. The clip of claim 16, wherein the extension is one of a pair of extensions spacing the spring finger portion from the remainder of the body, defining an opening between the pair of extensions.

18. The clip of claim 16, wherein the body further comprises a chassis attachment portion.

19. The clip of claim 18, wherein the clip further comprises a center wall spacing the chassis attachment portion from the spring finger portion.

20. The clip of claim 19, wherein the center wall meets each of the chassis attachment portion and the spring finger portion at a curved junction to define a first attachment region between the center wall and the spring finger portion and a second attachment region between the center wall and the chassis attachment portion.

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